

Spark Homes Repair Estimator

Contest Submission - Priyank Mandal

1. Most Interesting Design/UX Decision

Ditching full-DOM rebuilds for surgical updates. In a single-file app, the easy path is innerHTML teardowns on every checkbox click. But on a slow phone with 20+ room instances, that means rebuilding hundreds of DOM nodes just to flip one checkbox. So I made toggleItem, setQty, and setPrice update only the affected row class, line total, group badge, progress bar, section breakdown, and grand total. Nothing else. The DOM stays put. Scroll position, focus, and input state survive. It is the difference between butter and sandpaper, and it cost about 40 lines of code.

2. What is Broken or Fragile

Custom items are stored in an array and referenced by index. Delete one and everything after it shifts. It works because the event delegation reads live indices from data attributes, but it is one sneaky bug away from a meltdown. Also, the cost breakdown section only appears during a full render, so if you check the very first item of a project, the breakdown won't appear until you trigger a re-render by doing something else (adding a room, switching tabs, etc.). And photo capture on iOS sometimes shows a file picker instead of launching the camera. Apple, not me.

3. Creative Addition - Deal Analyzer

A repair estimate is useless without context. Knowing the repairs cost \$45K means nothing. Knowing they cost \$45K on a house with a \$200K ARV and \$120K purchase price, after \$8K in closing costs, \$7K in holding, and \$12K in selling costs, leaving a 4.2% margin? That is a decision. The Deal Analyzer computes full cost basis including closing costs, monthly holding, months held, and selling costs. It shows profit, ROI, and margin, all color-coded. All inputs persist per-project so you do not retype them when the page refreshes. This is the feature that makes the app actually useful in the field rather than just a fancy calculator.

4. What I Would Ship Next (Two More Days)

First, let agents tag photos to specific rooms or line items. The export would then show which photo documents which repair. Second, a JSON export/import for transferring estimates between devices without a backend. Third, a printable PDF scope-of-work summary. Contractors love paper.

5. Role of AI Tools

Claude Code was used for: generating the 108-item data model from the CSV, writing the IndexedDB wrapper, implementing the minimal ZIP and XLSX generators for offline export, debugging the service worker, writing the Playwright tests, and generating this writeup. The architecture, event delegation system, room/group data model, surgical DOM update layer, and deal analyzer logic were designed by me. AI was most useful for catching edge cases in the event delegation router and rapidly iterating the service worker caching strategy. It wrote maybe 30% of the code. I made sure the 70% that matters actually works.